

Episode 01 — Answer Key (Teacher Copy)

FOR TEACHER USE ONLY — Do
not distribute to students Grade 3 ·
Ages 8-9

The Mystery of the Magnetic Cat · NGSS 3-PS2-3 & 3-PS2-4

Pip's Discovery Lab ·
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SECTION 1 — VOCABULARY MATCHING

- Magnetic Field** → The invisible pushing-and-pulling zone around every magnet
- Ferromagnetic** → A metal that is naturally attracted to magnets (iron, nickel, or cobalt)
- Conductor** → A material that allows electricity or magnetic force to pass through it
- Poles** → The two ends of a magnet — North (N) and South (S)
- Repel** → To push away — what happens when the same magnetic poles face each other
- Attract** → To pull toward — what happens when opposite magnetic poles face each other

SECTION 2 — FILL IN THE BLANKS

1. Iron, Nickel, and Cobalt
2. I Need Chocolate — I = Iron, N = Nickel, C = Cobalt
3. skin (also accept: hand)
4. magnetic field
5. attract
6. repel
7. iron

SECTION 3 — TRUE OR FALSE

- All metals are magnetic. → **FALSE — Only iron, nickel, and cobalt are ferromagnetic.**
- A magnetic field can work through the air without touching an object. → **TRUE — Magnetic fields act at a distance.**
- A copper penny will stick to a horseshoe magnet. → **FALSE — Copper is not ferromagnetic.**
- A steel paper clip will stick to a magnet. → **TRUE — Steel contains iron, which is ferromagnetic.**
- North and South magnetic poles attract each other. → **TRUE — Opposite poles attract.**
- North and North magnetic poles attract each other. → **FALSE — Same poles repel.**
- Magnetic fields can pass through your hand to reach iron objects. → **TRUE — Confirmed in the episode with the paper clip demonstration.**
- Aluminum is one of the three ferromagnetic metals. → **FALSE — Aluminum is NOT ferromagnetic. The three are iron, nickel, and cobalt.**

SECTION 4 — SORT IT

MAGNETIC: Steel nail, Steel paper clip, Iron fridge door

NOT MAGNETIC: Copper penny, Plastic dinosaur, Aluminum soda can, Wooden block, Rubber band

SECTION 5 — SHORT ANSWER (Sample Answers)

1. A magnetic field is an invisible pushing-and-pulling zone around every magnet that can attract iron objects without touching them.
2. The soda can is made of aluminum. Aluminum is NOT a ferromagnetic metal — it does not contain iron, nickel, or cobalt — so it is not attracted to magnets.
3. I = Iron, N = Nickel, C = Cobalt. 'I Need Chocolate' helps us remember the three naturally magnetic metals.
4. When two same poles are put together, they REPEL — they push away from each other. You cannot force them to touch.
5. Pip discovered that Whiskers's metal collar tag was made of stainless steel, which contains iron — a ferromagnetic metal. That's why the magnet pulled the collar tag. The magnet was attracted to the iron in the tag, not to the cat himself.

NGSS ALIGNMENT

Standard	Description	Covered In
3-PS2-3	Ask questions to determine cause and effect relationships of electric or magnetic interactions between two objects not in contact with each other.	Sections 1, 2, 3, 5
3-PS2-4	Define a simple design problem that can be solved by applying scientific ideas about magnets.	Section 6 (Experiment)

TEACHER TIPS

Before watching: Ask students: 'What do you think makes something magnetic?' Record predictions on the board. Return to them after watching.

Pause points: Pause at 2:00 (after nail/penny tests) — ask 'What pattern do you notice?' Pause at 6:00 (after field demo) — ask 'What surprised you?'

After watching: Use Section 6 (Scavenger Hunt) as a take-home activity. Review results as a class next session.

Discussion: Ask: 'Why do you think only iron, nickel, and cobalt are magnetic but not other metals like copper or gold?' (No wrong answers — encourage curiosity.)

Extension: Challenge early finishers to design a 'magnetic sorter' machine using what they learned. What would it look like? What would it separate?

Science verification: All facts in this episode and worksheet are verified against NGSS, Britannica Kids, ACS (American Chemical Society), and Science Trek.

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